

AJAY KRISHNA DEVULAPALLY

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PROFESSIONAL SUMMARY

Machine Learning Engineer and M.S. Computer Science student at Penn State with experience building ML workflows, model evaluation pipelines, data-driven research systems, and systems-oriented software. Skilled in Python, C/C++, Java, SQL, PyTorch, TensorFlow, Keras, model evaluation, benchmarking, data pipelines, object-oriented design, Git-based development, and scalable ML experimentation.

EDUCATION

The Pennsylvania State University

M.S. in Computer Science, Thesis Track

University Park, PA

Aug 2024 – Aug 2026

CVR College of Engineering

B.Tech. in Computer Science and Engineering

Hyderabad, India

2019 – 2023

EXPERIENCE

Graduate Research Assistant

Microsystems Design Lab, Penn State | SRC

Aug 2024 – Present

University Park, PA

- Conduct SRC-supported research under **Prof. Vijaykrishnan Narayanan** on AI/ML-driven optimization for microsystems and combinatorial design problems.
- Reduced wirelength by up to 27% and achieved 58× faster time-to-solution by building an RL-based 2.5D chiplet placement system using PPO, imitation learning, routing optimization, and thermal simulation.
- Built Python/PyTorch experiment pipelines for simulation, benchmarking, model evaluation, failure analysis, and metric-driven iteration across constrained optimization problems.
- Developing GNN-based freeze-mask prediction for oscillator-based Ising-machine optimization, using node- and graph-level features to reduce runtime while preserving solution quality on unseen graphs.

AI Intern

Smartknower

Mar 2021 – Apr 2021

Bangalore, India

- Built ANN-based and transfer-learning image classification models, achieving 98.3% accuracy on MNIST.
- Developed preprocessing, validation, evaluation, and deployment workflows using Python, OpenCV, Pandas, and Streamlit.

RESEARCH

TWIRLeT-2.5D – First Author

[*TCAD 2026*] *Ajaykrishna Devulapally, Varun Parekh, Anusha Devulapally, Mirco Sciuilli, Srivatsa Rangchar Srinivasa, Vijaykrishnan Narayanan, “TWIRLeT-2.5D: Thermal Wirelength Integrated Reinforcement Learning for 2.5D Chiplet Placement.”*

- Designed an end-to-end reinforcement-learning system for thermally aware 2.5D chiplet placement using imitation learning, PPO, MILP-based routing, and thermal simulation.
- Developed evaluation pipelines across industrial-scale environments, reducing wirelength by up to 27% and achieving 58× faster time to solution than simulated annealing.

PROJECTS

3D Human Pose Forecasting using SMPL & VPoser

- Built GRU- and Transformer-based PyTorch models to forecast 3D human motion sequences from latent pose representations.
- Evaluated generated motion using MPJPE, PA-MPJPE, and MPVPE, strengthening experience in sequence modeling, computer vision, representation learning, and model evaluation.

DistilGPT2 Policy Optimization and Evaluation

- Fine-tuned and evaluated DistilGPT2 on task-specific text data, building reproducible workflows for preprocessing, training, checkpointing, and evaluation using PyTorch and Hugging Face Transformers.
- Compared baseline and policy-optimization experiments using reward and evaluation metrics, with failure analysis focused on generation quality, training stability, repetition, and reward-design trade-offs.

Skin Cancer Detection using Hybrid Deep Learning

- Achieved 90.625% classification accuracy by building a medical-image classification workflow using VGG16, ResNet, and CapsNet-style architectures.
- Implemented image preprocessing, transfer learning, model training, ensemble-style prediction, and quantitative evaluation workflows.

SKILLS

Programming: Python, C++, C, SQL, Java

ML / AI: PyTorch, TensorFlow, Keras, Scikit-learn, Hugging Face Transformers, Reinforcement Learning, Imitation Learning, GNNs, Transformers, CNNs, RNNs, Computer Vision, NLP

ML Engineering: Dataset preparation, model training and fine-tuning, reward design, experiment pipelines, benchmarking, model evaluation, error analysis, checkpointing, regression testing, metric-driven iteration

Systems / Data: Linux, Git, NumPy, Pandas, OpenCV, Streamlit, data structures, algorithms, performance analysis, ScaleSim

Certificates: Artificial Intelligence – Smartknower (2021), Database Programming with SQL – Oracle (2022)